How Much of Wearable Activity-Recognition Accuracy Is the Network? A Pre-Registered Variance Decomposition Under Person-Disjoint Evaluation

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***Abstract*—Wearable inertial activity-recognition systems are compared by their macro-F1 on people unseen in training, but how much of that figure reflects the network and how much the people who happen to be tested has not been measured. We decompose the variance of stored runs of one fixed person-disjoint protocol (WHAR Arena v1) with a linear mixed model whose confirmatory test was pre-registered. On 7 held-out datasets (828 runs of four networks with three training seeds), the group of test subjects accounts for 82 % of the within-dataset variance, 8.19 times (97.5 % interval 2.85–16.90) the variance of all network-related terms together, and the pre-registered criterion was met. The ordering also holds with a metric free of per-split ceilings and on the development datasets, where its interval includes 1 for that metric. It reverses across the benchmark's 17 heterogeneous published architectures, where the network dominates. Between comparable networks, the subjects a wearable study records bound what an accuracy difference can show.**

***Index Terms*—Wearable sensors, inertial sensors, human activity recognition, person-disjoint evaluation, variance components, mixed models.**

I. Introduction

Wearable inertial sensing systems for human activity recognition (HAR) are designed and chosen on one number: macro-F1 on subjects held out from training [1]. Evaluating on unseen people removes the optimism of random cross-validation [2], and how people are partitioned between training and test data changes the reported accuracy [3]. People move differently [4] and devices sense differently [5], so a test set of a few subjects carries its own variation. WHAR Arena [6] fixed one person-disjoint protocol across 30 datasets and found the state of the art distributed over architectures rather than held by one; it reports one training run per split, so it cannot separate the effect of who is tested from the training seed. Machine-learning benchmarks are known to depend on such sources of variation [7], but their sizes in wearable HAR are unknown.

This letter measures them. Under one fixed protocol it decomposes the variance of macro-F1 into the group of test subjects, the network, their interactions and the training seed; it tests the ordering of these terms with a pre-registered analysis on held-out datasets; it checks the result with a ceiling-free metric, development data and the benchmark's independent architectures; and it reports, as exploratory, which design choices and which sensing properties of a dataset go with large accuracy differences.

II. Methods

A. Protocol and stored runs

All runs follow WHAR Arena v1 with its paper-era code [6], [8]: 3-s windows with 50 % overlap at the native sampling rate; $k = 10$ subject groups, each split testing on one group (1 to 4 subjects) that never appears in training or validation; validation from 20 % of the training subjects' windows; a z-score fitted on the training split; macro-F1 over all classes, with classes absent from a test split counted as 0. No network was trained for this letter: we analyse stored runs and their checkpoints. Three sets enter (Table I): S1, ten networks that share one skeleton and differ in single components (with and without temporal memory, e.g. S4D [9], [10] and Mamba [11], or window statistics), on ten development datasets with three seeds; S2, four of these networks on eight datasets never used for this question, with three seeds; S3, the benchmark's 17 published models (deep networks such as DeepConvLSTM [12], [13], and Random Forest [14], SVM and k-NN on tsfresh features [15]) re-run with the same code on four datasets, one run per split. A pre-registered inclusion rule removed datasets whose stored runs disagree on split sizes, because their split indices do not name the same subjects across runs (one dataset each in S1 and S2).

B. Variance model and pre-registered test

For run (m, d, s, r) of network m on split s of dataset d with seed r,

$$y_{mdsr} = \mu_d + P_{ds} + M_m + (MD)_{md} + (MP)_{mds} + e_{mdsr}, \quad (1)$$

with μ_d fixed and the other terms independent zero-mean random effects, fitted by REML [16] in statsmodels [17]. The estimand is $R = \sigma_P^2/(\sigma_M^2 + \sigma_{MD}^2 + \sigma_{MP}^2)$, the variance of the test-subject group over all network-related variance; π_P is σ_P^2's share of the within-dataset variance. Intervals are percentile 97.5 % intervals from 1000 cluster-bootstrap resamples that draw splits within datasets [18]. Before any variance analysis of S2, we registered H1 ($R > 1$ on S2), with success if the interval's lower bound exceeds 1 and failure if $R \leq 1$, sealed the analysis code by hash, and logged S2's single read. S2's per-dataset means were known from an earlier study; its variance components were not. For S3 (one run per split) the interaction and the seed cannot be separated, so the conservative $R' = \sigma_P^2/(\sigma_M^2 + \sigma_{MD}^2 + \sigma_e^2)$ is reported.

C. Exploratory analyses and verification

Component contrasts on S1 pair two networks by split, average the seeds, and summarise over datasets with a t-interval and a Wilcoxon test [19], [20]. Over the benchmark's 30 datasets we regress the published single-run results on standardised dataset properties (ordinary least squares); these associations are correlational, and their p-values are not corrected for multiplicity. For verification, the benchmark's Random Forest, re-run with our code, reproduces its published macro-F1 on four datasets (e.g. 84.69 against 84.69 on UCI-HAR); test predictions were regenerated from 408 checkpoints, 373 of which reproduced the stored metrics exactly and all within two test windows; an independent REML implementation and negative controls (e.g. permuting network labels within cells) confirmed the analysis. The regenerated runs include 360 of the 828 confirmatory runs (the datasets cached on our cluster); the others, and one development dataset (DSADS) whose cache on the cluster does not reproduce the stored runs, enter on their stored records.

III. Results

A. Confirmatory test

On S2 the test-subject group dominates (Fig. 1, Table I): $\sigma_P^2 = 127.7$ (F1 points^2) against $\sigma_M^2 = 4.8$, $\sigma_{MD}^2 = 4.0$ and $\sigma_{MP}^2 = 6.9$,

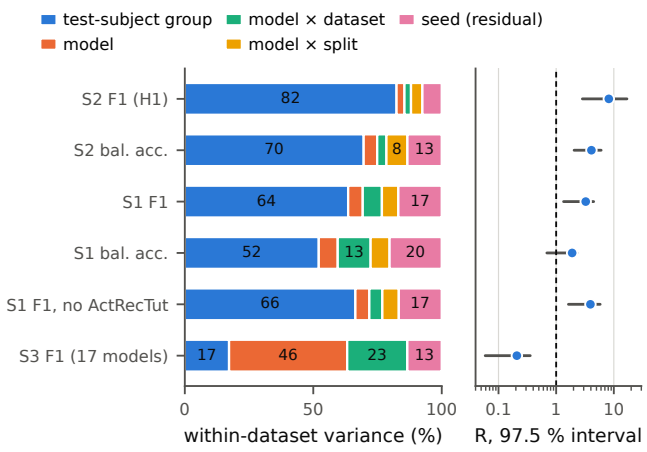


Fig. 1. Where the within-dataset variance of the stored runs lies. Left: shares by source (numbers in %). Right: R with its 97.5 % bootstrap interval; the dashed line is the pre-registered threshold. R exceeds 1 in every analysis of the shared-skeleton networks (on S1 with balanced accuracy its interval includes 1), and the network dominates across the 17 heterogeneous published architectures (S3, conservative R').

TABLE I
VARIANCE COMPONENTS (F1 POINTS^2) AND R [97.5 % INTERVAL]

Analysis	n	σ^2_P	σ^2_M	σ^2_{MD}	σ^2_{MP}	σ^2_e	R [97.5 % int.]
S2 F1 (H1)	828	127.7	4.8	4.0	6.9	11.7	8.19 [2.85, 16.90]
S2 bal. acc.	828	86.3	6.5	4.4	10.1	16.6	4.09 [2.05, 6.03]
S1 F1	2160	66.2	5.9	7.8	6.7	17.6	3.26 [1.34, 4.47]
S1 bal. acc.	2160	39.2	5.5	9.6	5.5	15.3	1.89 [0.69, 2.29]
S1 F1, no ActRecTut	2100	67.4	5.5	5.1	6.5	17.0	3.93 [1.63, 5.80]
S3 F1 (17 models)	510	37.5	100.2	51.0	–	29.2	0.21 [0.06, 0.36]

with the seed at $\sigma^2_e = 11.7$. Hence $R = 8.19$ [2.85, 16.90] and $\pi_P = 82\,\%$ [64, 89]: the pre-registered decision is *success*, and the secondary criterion (π_P lower bound $\geq 40\,\%$) is met.

B. Sensitivity analyses

Balanced accuracy, which averages recall over the classes present and so has no per-split ceiling, gives $R = 4.09$ [2.05, 6.03] on S2: ceilings enlarge the subject share but do not create it. On the development set S1 the ordering is the same with macro-F1 ($R = 3.26$ [1.34, 4.47]), also without its two-subject dataset ActRecTut ($R = 3.93$); with balanced accuracy the point estimate stays above 1 but the interval does not exclude it ($R = 1.89$ [0.69, 2.29]). Across the benchmark's 17 published architectures it reverses: $R' = 0.21$ [0.06, 0.36], with $\sigma^2_M = 100.2$ against $\sigma^2_P = 37.5$. These architectures range from k-NN to attention networks; their differences are far larger than those between the shared-skeleton networks.

C. Exploratory findings

Within S1, the one design choice worth several points is temporal memory (Fig. 2a): S4D gains +5.79 [+2.30, +9.27] points over the same network without memory (9 of 9 datasets), while the type of memory moves macro-F1 by at most a few points (Mamba − S4D: −2.45 [−4.40, −0.50]). Over the 30 datasets, a larger channel count goes with a larger margin of the feature-based Random Forest over the best deep network (+7.1 points per standard deviation of $\log_2$ channels, $p < 0.001$, other properties held fixed; Fig. 2b), and fewer training subjects go with a larger gap between the best and the median deep network (−1.5 per standard deviation, $p = 0.004$). No property predicts how hard a dataset is ($R^2 = 0.03$).

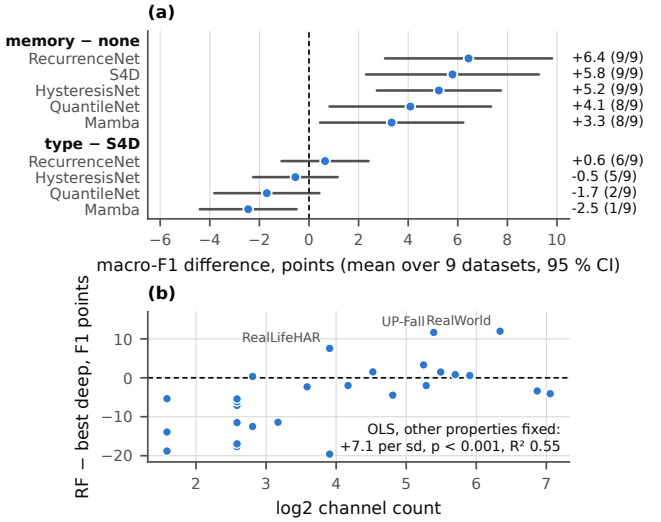


Fig. 2. Exploratory. (a) Temporal memory against none, and memory types against S4D, over the development datasets whose runs pair by subject (mean and 95 % CI; datasets won). (b) Over the benchmark's 30 datasets (published single runs), the Random Forest's margin over the best deep network rises with the number of sensor channels.

IV. Discussion and Limitations

Here "comparable" networks share one design and differ in single components, so that their dataset means differ by a few points. For such networks, who is tested decides more of a reported macro-F1 than which network is used. The variance between test-subject groups governs the absolute level, and with it every comparison made across studies, splits or subject groups. A comparison paired by split cancels it, but is then limited by the network × split interaction and the seed, which on S2 are as large as the network terms themselves (Table I). For a wearable sensing study this gives three rules. Record enough subjects to estimate the spread across test-subject groups, and report it. Compare networks paired on the same splits, never by means over different subjects. Average several training seeds. The reversal on S3 bounds the claim: a weak baseline is told apart from a strong network easily, and the subject effect matters when choosing between designs that are already competitive. The exploratory associations point at the sensing system: with many channels, hand-crafted features stay competitive, and with few recorded subjects the choice of network matters more.

The analysis covers one protocol, and three seeds per run. The S1 and S2 networks share one skeleton, so their components are controlled but not independent designs. S2's means were known before the registration. The 30-dataset results re-analyse published single runs and are correlational, and subjects and channels co-vary across datasets. No network was trained for this letter, so no label-permutation control was run; the splits are subject-disjoint by construction and the normalisation is fitted on the training split, which we verified, but a leak through another path would not have been detected.

V. Conclusion

Under a fixed person-disjoint protocol, who is tested accounts for most of the variance of macro-F1 among comparable wearable HAR networks, as a pre-registered test on held-out datasets confirmed; across heterogeneous architectures the network dominates. Studies of wearable sensing systems should size and report their test subjects accordingly.

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